



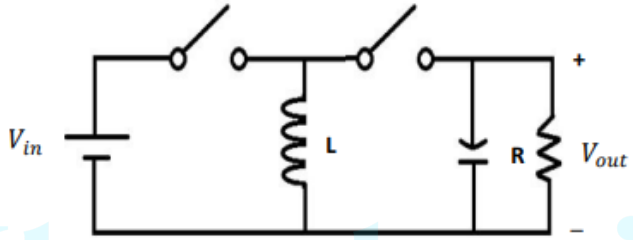
KTU NOTES

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NOTIFICATIONS | SOLVED QUESTION PAPER**

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4. The switches in the figure are operated alternatively, each switch being on for one cycle. Determine the relationship between V_{in} and V_{out} .



5. A step up chopper has input voltage of 120V and output voltage of 360 V. If the on time of the thyristor chopper is 100 μ s, Compute the pulse width of output voltage.
6. With the help of circuit diagram and waveform explain the operation of a step up chopper. Also derive the equation of output voltage.
7. Draw the circuit of step up chopper and explain its working.
8. For a type A chopper, dc source voltage is 230 V, load resistance 10 Ω , dc load current is 2V and duty cycle 0.4. Calculate average and RMS value of output voltage and chopper efficiency
9. Draw the circuit of buck boost converter and explain its working.
10. Explain different types of chopper
11. A step up chopper has input voltage of 220V and output voltage of 400 V. If the on time of the switch is 100 μ s, Compute the pulse width of output voltage.
12. For a dc-dc buck-boost converter with a dc input voltage of 50V and output voltage of 100V, calculate (i) duty cycle (ii) value of inductor if inductor ripple current $\Delta I = 1$ A and switching frequency is 10kHz
13. Draw the waveform of inductor voltage of a boost dc-dc converter and output voltage for output dc voltage in terms of input voltage and duty cycle.
14. In a step down chopper the dc input voltage is of 100V. The MOSFET switch has a switching frequency of 2kHz. Find the duty cycle and average dc output voltage if the period of switch is 0.2ms
- 15.

4. Explain with circuit diagram and waveforms, the working of Buck regulator in continuous current mode. Obtain expressions for inductance and capacitance.
5. Explain the working of Two quadrant (Class C) chopper, with relevant waveforms.
6. Derive the expression for output voltage of a Buck-Boost regulator, show the waveforms.
7. How four-quadrant operation is achieved in a Type E Chopper? Explain with circuit diagram.
8. With circuit diagram and waveforms, describe the operation of a buck-boost converter. Derive expressions for output dc voltage and the design equations for filter inductor and capacitor.
9. For a dc-dc buck-boost converter with a dc input voltage of 50V and output voltage of 100V, calculate (i) duty cycle (ii) value of inductor if inductor ripple current $\Delta I_L = 1A$ and switching frequency is 10kHz.
10. With circuit diagram and waveforms, describe the operation of a buck-boost converter. Derive expressions for output dc voltage and the design equations for filter inductor and capacitor.
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